California Economic Trends

CPI, Housing, Minimum Wage & U.S. GDP

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Final Course Project Presentation

Objective & Hypothesis

- Integrate related economic and sensor datasets using Year as the common time variable.
- Examine whether changes in California CPI move alongside median home prices, housing CPI, minimum wage, and broader U.S. GDP trends.
- Use exploratory analysis and visualization to identify patterns rather than claim direct causation.

Research focus
2010–2025

Key comparison
California CPI ↔ Housing

Data Integrity & Data Validation

Data Integrity

- Checked column names and data types before analysis.
- Reviewed missing and duplicate observations before integration.
- Kept CPI and housing measures numeric for comparison.

Data Validation

- Validated Year as the common key across the CPI and housing tables.
- Confirmed California CPI Index and Housing CPI are numeric fields.
- Used the shared 2010–2025 period to support year-by-year integration.

Validation before merge

CPI table → Year key ← Housing table

This protects the integrity of the joined table and reduces mismatched observations.

Data Collection & Key Variables

Primary dataset
1,120 rows
2010–2025

Housing dataset
48 rows
2010–2025

GDP dataset
United States
annual GDP

- Primary variables: California CPI Index, CPI Annual Change %, California Minimum Wage, Industry, Occupation and Region.
- Housing variables: Housing CPI, Overall CPI, Median Rent, Median Home Price and Mortgage Rate.
- GDP data is reshaped from year columns into Year and GDP, then filtered to the United States.

Data Wrangling to Impact and Segmentation

- To form a better analysis, Housing CPI and California CPI is updated and read to visualization of datatype, as well as, execution to numeric values.
- Understood data tendencies to find statistical output and relational output on both keys ['California CPI' and 'Median Home Price'] using merge function.
- These insights showed summary data with unique identifiers such as ['Year'] to replace any data contingency.

Applied the Year key to join the California CPI Index and Housing CPI tables.

CPI INDEX — df.dtypes

Year	int64
Industry	str
Occupation	str
Region	str
Employment Type	str
California Minimum Wage	float64
California CPI Index	float64
CPI Annual Change %	float64

HOUSING CPI — df.dtypes

Year	int64
Industry	str
Housing CPI	float64
Overall CPI	float64
Median Rent	int64
Median Home Price	int64
Mortgage Rate	float64

Exploratory Data Analysis

- Prepared the merged CPI and housing data for exploratory analysis.
- Created a MultiIndex using Year, Industry, California Minimum Wage, and Median Home Price.
- Sorted the index in descending order so the latest observations appear first.

MultiIndex focus used for analysis

```
user_focus = user_analysis.set_index(
    ["Year", "Industry_x",
     "California Minimum Wage",
     "Median Home Price"]
).sort_index(ascending=False)

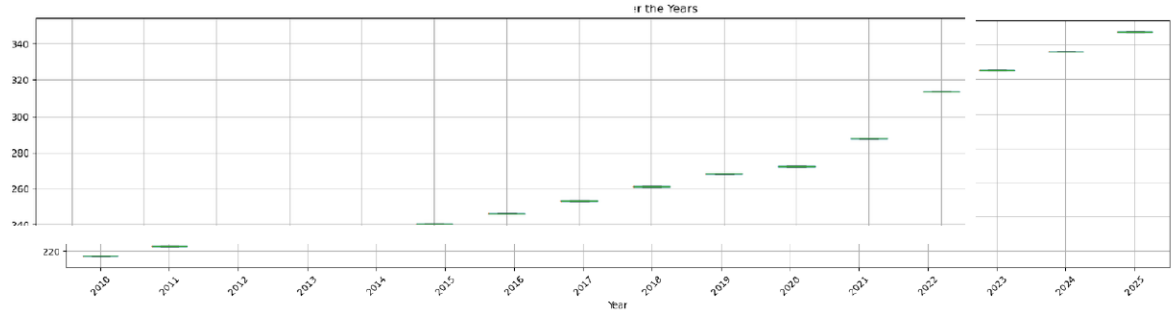
user_focus.iloc[:10]
```

Limitation
Top home-price values do not represent every California housing market.

Interpretation
Correlation or similar movement does not prove causation.

Distribution Analysis

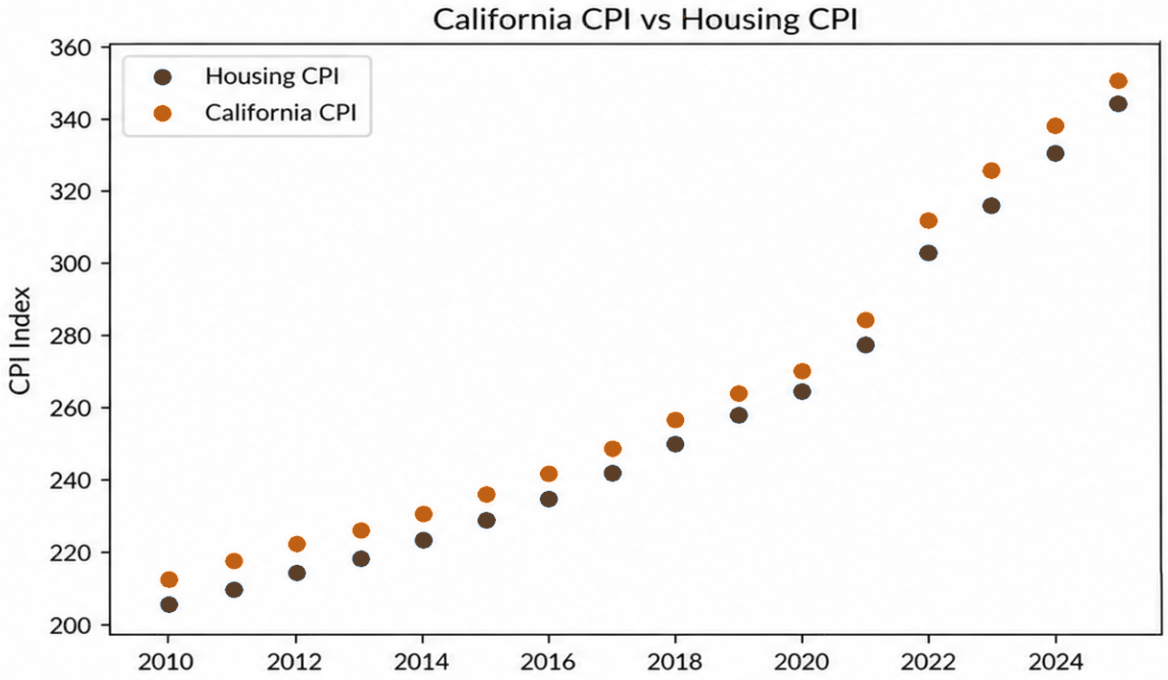
:alifornia CPI Index by Year



- The CPI distribution shifts upward over the 2010–2025 period.
- Later years show higher CPI levels than earlier years.
- Distribution data shows how time and value affect California CPI within a specific timeline and across selected factors.

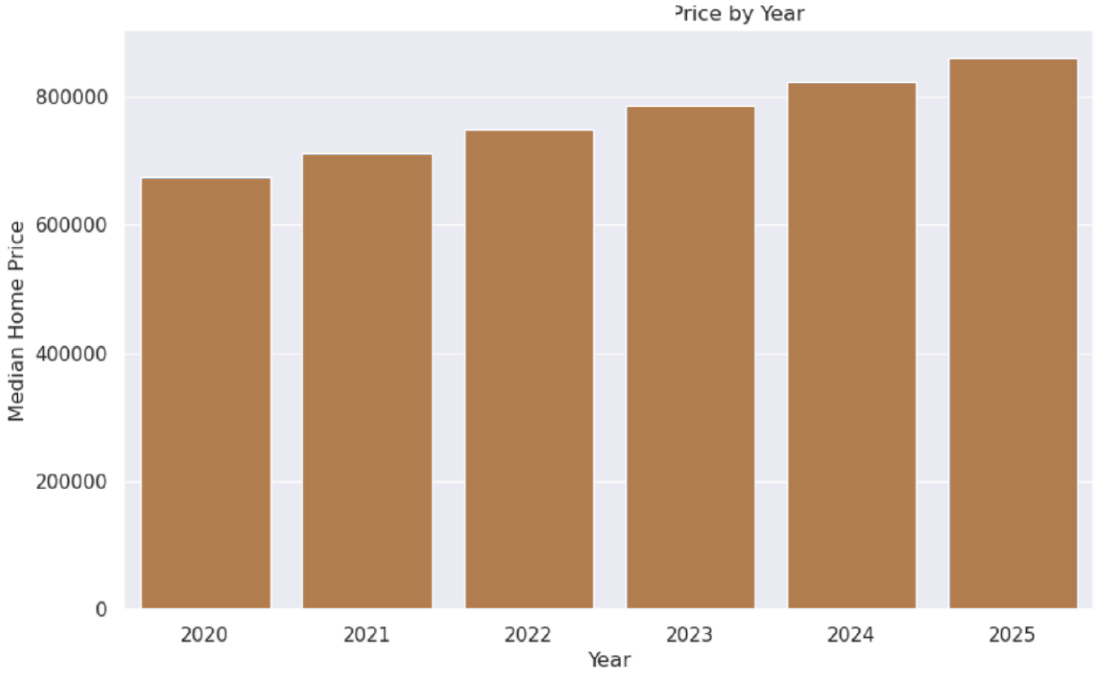
Relational Analysis

California CPI vs. Housing CPI



- Both CPI measures generally rise over time.
- Housing CPI and California CPI show similar directional movement across the observed years.
- The comparison suggests a positive relationship worth evaluating, while remaining descriptive rather than causal.

Recent Housing Trend



- The notebook focuses on recent median home prices from 2020 through 2025.
- The bar chart highlights the upward movement in the median home-price series.
- Housing prices can move differently from broad CPI because housing is influenced by mortgage rates, supply, location and other market conditions.

Key Findings

1. California CPI and Housing CPI both trend upward across the study period.
2. Recent median home prices show strong upward movement, supporting the housing-cost component of the economic trend analysis.
3. GDP provides broader economic context, but the notebook treats it as a comparison variable rather than evidence of a direct effect.

Conclusion, Limitations & Next Steps

- Conclusion: California CPI, housing CPI and median home prices show measurable upward trends over the study period.
- Limitation: statewide CPI and median home prices summarize broad conditions and may hide regional or industry-level differences.
- Limitation: the analysis identifies association and trend, not a causal relationship between CPI, wages, housing prices and GDP.
- Next step: expand the analysis with regional housing data, additional years, and formal correlation or regression measures.

Presentation designed for a high-level briefing of ~10 minutes or less.